

Upgrade NX3232c Cluster Switch

Need Help?

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This process is the high level steps to perform a Nexus 3232x Switch NX-OS and/or RCFUpgrade, these are performed by NetApp Managed Services as part of the standard operational support function and will be linked to any change request for performing Switch upgrades. This is so that the technical steps can be reviewed by the approvers necessary.

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Before You Begin

Before you begin, you must meet these requirements:

- Ensure you have connectivity/access to both Cluster and Cluster switches
- Have reviewed the OnTap upgrade advisor for details on Switch dependency etc (if performing the switch upgrade prior to the OnTap upgrade)
- Reviewed the interdependencies and compatibility
 - Hardware Universe <https://hwu.netapp.com/>
 - Interoperability Matrix <http://mysupport.netapp.com/matrix>
- Taken a copy of the Cluster switch **running config** and stored it in case of roll back if performing an NX-OS or RCF upgrade
- Taken a copy of the **RCF file** and stored it in case of roll back if performing an NX-OS or RCF upgrade
- Verify all Switch cabling is done correctly as detailed in the config advisor report or via the Cluster
- You must have checked or set your desired boot configuration in the RCF to reflect the desired boot images if you are installing only NX-OS and keeping your current RCF version.
- If you need to change the boot configuration to reflect the current boot images, you must do so before reapplying the RCF so that the correct version is initiated on future reboots.

You **must have a console connection** to the switch when **installing or upgrading the RCF file** - the switch will be wiped, so console access is a MUST !

This process is focused on mainly NX-OS upgrades with the necessary pre/post steps, as they are likely to be performed almost as regularly as an OnTap upgrade, so the format of this wiki article has been laid out as such - however, in some circumstances, an RCF file upgrade maybe required, so although the steps to complete this have been documented **last**, it is best practice to perform the RCF upgrade **first**, as essentially once this is applied to a working Cluster switch, it will bring the Cluster switch back to normal operation.

PRE Check Steps to be carried out from the Cluster and the Switches

A recently captured config advisor report or output of NetApp docs for both Cluster and the attached Switches is required to confirm there are no cabling issues (attached to the change as confirmation) - **each NODE needs to be connected to Switch 1 and Switch 2 NOT the SAME switch**

All of the output may need to be captured for Post Implementation Review, so it's worth saving the console sessions in case they are needed for troubleshooting any unforeseen issues.

STEP	COMMAND / DETAILS
Complete a config advisor report	Check the Cluster Switch and Array details for any issues
To be performed on Cluster Switch:	

Check version of RCF file installed on the Cluster Switch (the RCF version needs to support the current OnTap version of OnTap or the version being upgraded to if the Switch is being upgraded prior to OnTap) - for example: • Cluster RCF_v1.0 supports OnTap 9.3 to 9.8 • Cluster-HA RCF 1.6 supports OnTap 9.6 to 9.9.1 • Cluster-HA RCF 1.8 supports OnTap 9.6 to 9.10.1 • NetApp Support Site - Cisco Ethernet Switch	A pre-requisite before an NX-OS or RCF Upgrade <pre>show running-config</pre> Check the output, it's normally in the banner: <pre>***** * * NetApp Reference Configuration File (RCF) * * Switch : Cisco Nexus 3232C * Filename : Nexus_3232C_RCF_v1.6-Cluster-HA.txt * Date : Oct-20-2020 * Version : v1.6 * * Port Usage : Non breakout configuration * Ports 1-30: 40/100GbE Intra-Cluster/HA Ports, int e1/1-30 * Ports 31-32: Intra-Cluster ISL Ports, int e1/31-32 * Ports 33-34: 10GbE Intra-Cluster 10GbE Ports, int e1/33-34 * * IMPORTANT NOTES * - Load Nexus_3232C_RCF_v1.6-Cluster-HA-Breakout.txt for breakout config *****</pre> You can save the current rcf file from admin host (lp-nj1-gipat001) and save the output direct to a file on your homedir: <pre>ssh admin@[switch] show file {rcf.txt} > {date}_{switch}_{rcf.txt}.txt</pre> <pre>lp-nj1-gipat001:/home/scargill \$ ssh admin@10.76.5.211 show file Nexus_3232C_RCF_v1.6-Cluster-HA.txt > 20211230_cs_oth_sy5_prd1_sw2_Nexus_3232C_RCF_v1.6-Cluster-HA.txt</pre>
Check / Change Console timeout settings (default is 30 mins, a previous NX-OS copy of a 2Gb file took 36 minutes)	To be run in admin mode (conf t) <pre>conf t line vty exec-timeout 60 exit (ctrl-z)</pre> Once done, this can be seen in the running-config: <pre>line vty exec-timeout 60 boot nxos bootflash:/nxos.9.3.7.bin</pre>
Create a backup copy of the running config	You can save the running-config from admin host (lp-nj1-gipat001) and save the output direct to a file on your homedir: <pre>ssh admin@[switch] show running-config > {date}_{switch}_config.txt</pre> <pre>lp-nj1-gipat001:/home/scargill \$ lp-nj1-gipat001:/home/scargill \$ ssh admin@10.76.5.210 show running-config > 20211230_cs_oth_sy5_prd1_sw1_config.txt</pre>
Check status of Fan's and PSU's etc	Ensure there are no issues <pre>show environment show inventory</pre>

Check status of modules	Ensure all modules are available show module <pre>cs_oth_sy5_prd1_sw1# show module Mod Ports Module-Type Model Status ----- 1 34 32x40/100G QSFP28 2x10G SFP+ Ethernet N3K-C3232C active *</pre>
Check switch can see Cluster ports	Ensure there are 2 interfaces connected to diverse switches show cdp neighbors <pre>Device-ID Local Intrfce Hldtme Capability Platform Port ID CS_OTH_SY5_PRD1-01 Eth1/7 174 H AFF-A800 e0a CS_OTH_SY5_PRD1-01 Eth1/8 174 H AFF-A800 e1a CS_OTH_SY5_PRD1-03 Eth1/9 157 H AFF-A800 e0a CS_OTH_SY5_PRD1-03 Eth1/10 157 H AFF-A800 e1a cs_oth_sy5_prd1_sw2 (FOC24261V6C) Eth1/31 176 R S I s N3K-C3232C Eth1/31 cs_oth_sy5_prd1_sw2 (FOC24261V6C) Eth1/32 176 R S I s N3K-C3232C Eth1/32</pre>
Check status of all interfaces	All the interfaces should be up and connected show int brief grep up <pre>mgmt0 -- up 10.76.5.210 Eth1/7 1 eth trunk up none Eth1/8 1 eth trunk up none Eth1/9 1 eth trunk up none</pre>
Check status of all ports (inter-cluster)	Confirm the port channel is up show port-channel summary <pre>----- Group Port-Channel Type Protocol Member Ports ----- 1 Pol (SU) Eth LACP Eth1/31 (P) Eth1/32 (P)</pre>

Confirm version of NX-OS	Ensure the switch version is the expected version prior to an upgrade to a new version (where the change in NX-OS version will become be reflected in bootflash) <pre>show version</pre> <pre> Software BIOS: version 08.38 NXOS: version 9.3(7) BIOS compile time: 05/29/2020 NXOS image file is: bootflash:///nxos.9.3.7.bin NXOS compile time: 3/10/2021 3:00:00 [03/10/2021 13:39:29] Hardware cisco Nexus3000 C3232C Chassis (Nexus 9000 Series) Intel(R) Xeon(R) CPU E5-2403 v2 @ 1.80GHz with 8154108 kB of memory. Processor Board ID FOC24261V43 </pre>
To be performed on NetApp Cluster:	
Check monitoring is enabled	These should be set to true (if they are not, the steps to fix this are covered further down) <pre>system cluster-switch show -fields is-monitoring-enabled-operational,reason</pre> <pre> device reason is-monitoring-enabled-operational ----- cs_oth_sy5_prdl_sw1(FOC24261V43) None true cs_oth_sy5_prdl_sw2(FOC24261V6C) None true 2 entries were displayed. </pre>
Check that there are no storage switch or cabling issues	This should already have been done as part of the config advisor check, but can be done at the console level (although once reported, these need to be manually cleared, so isn't a true reflection of current state) <pre>system health alert show -instance</pre> <pre> CS_OTH_SY5_PRDL1:> system health alert show -instance Node: CS_OTH_SY5_PRDL1-03 Monitor: ethernet-switch Alert ID: ClusterSwitchConnectivity_Alert Alerting Resource: CS_OTH_SY5_PRDL1-01 Subsystem: Switch-Health Indication Time: Tue Dec 14 11:35:10 2021 Perceived Severity: Major Probable Cause: Configuration_error Description: One or more nodes are not connected to both cluster switches. </pre> You should see the following: <i>There are no entries matching your query</i> (the above output was manually manipulated to show you what a storage switch or cabling issue would look like if you came across one)
Send Maintenance ASUP from the Cluster	If we are not putting the Cluster in maintenance mode in AIQUM, we should send an ASUP so no cases are auto-generated <pre>system node autosupport invoke -node * -type all -message MAINT=x (where x = hours)</pre>

Check the Cluster Switch Version	Any issues here may mean a problem with SNMP <pre>system cluster-switch show</pre> <pre>cs_oth_sy5_prd1_sw2 (FOC24261V6C) cluster-network 10.76.5.211 NX3232C Serial Number: FOC24261V6C Is Monitored: false Reason: Invalid SNMP Settings Software Version: Cisco Nexus Operating System (NX-OS) Software, Version 9.3(7) Version Source: CDP</pre> Check SNMP is configured correctly if there are issues <pre>system cluster-switch show -fields snmp-version,community-or-username</pre> <pre>device snmp-version community-or-username ----- - cs_oth_sy5_prd1_sw1 (FOC24261V43) SNMPv2c cshml! cs_oth_sy5_prd1_sw2 (FOC24261V6C) SNMPv2c cshml! 2 entries were displayed.</pre>
Display how many cluster interconnect interfaces are configured in each node for each cluster interconnect switch	<pre>network device-discovery show -protocol cdp</pre>
Display how many Cluster interfaces are specific to the Cluster Switch	Each IC LIF on the Cluster should be diversely cabled to the Cluster switches: <pre>network device-discovery show -platform N3K-C3232C</pre> <pre>Node/ Local Discovered Protocol Port Device (LLDP: ChassisID) Interface Platform ----- CS_OTH_SY5_PRD1-04/cdp e0a cs_oth_sy5_prd1_sw1 (FOC24261V43) e0b CS_OTH_SY5_PRD1-03 Ethernet1/10 N3K-C3232C e1a cs_oth_sy5_prd1_sw2 (FOC24261V6C) e1b CS_OTH_SY5_PRD1-03 Ethernet1/10 N3K-C3232C AFF-A800 AFF-A800</pre>

Confirm all interfaces are in the same Broadcast Domain

Check all the Cluster interfaces are up in the Cluster broadcast domain

```
broadcast-domain show
```

```
IPspace Broadcast
Name      Domain Name  MTU  Port List
-----
Cluster Cluster    9000
CS_OTH_SY5_PRD1-04:e0a  complete
CS_OTH_SY5_PRD1-04:e1a  complete
CS_OTH_SY5_PRD1-02:e0a  complete
CS_OTH_SY5_PRD1-02:e1a  complete
CS_OTH_SY5_PRD1-03:e0a  complete
CS_OTH_SY5_PRD1-03:e1a  complete
CS_OTH_SY5_PRD1-01:e0a  complete
CS_OTH_SY5_PRD1-01:e1a  complete
```

Display the network port attributes

A further check for the Cluster LIF's

```
network port show -node * -ipspace Cluster
```

```
Node: CS_OTH_SY5_PRD1-01
Port      IPspace      Broadcast Domain Link MTU  Speed(Mbps) Health
-----
e0a       Cluster      Cluster      up   9000  auto/100000 healthy
e1a       Cluster      Cluster      up   9000  auto/100000 healthy
```

Display information about the LIFs

A further check for the Cluster LIF's

```
network interface show -vserver Cluster
```

Ping the remote cluster LIF

To make sure there is connectivity within the Cluster before progressing with either an NX-OS or RCF upgrade

```
cluster ping-cluster -node {node-name} -use-sitelist true
```

```
Host is CS_OTH_SY5_PRD1-01
Getting addresses from sitelist...
Local = 169.254.36.40 169.254.25.67
Remote = 169.254.184.165 169.254.155.235 169.254.93.231
Cluster Vserver Id = 4294967293
Ping status:
.....
Basic connectivity succeeds on 12 path(s)
Basic connectivity fails on 0 path(s)
.....
Detected 9000 byte MTU on 12 path(s):
  Local 169.254.25.67 to Remote 169.254.155.235
  Local 169.254.25.67 to Remote 169.254.157.109
  Local 169.254.25.67 to Remote 169.254.181.246
  Local 169.254.25.67 to Remote 169.254.184.165
  Local 169.254.25.67 to Remote 169.254.241.217
  Local 169.254.25.67 to Remote 169.254.93.231
  Local 169.254.36.40 to Remote 169.254.155.235
  Local 169.254.36.40 to Remote 169.254.157.109
  Local 169.254.36.40 to Remote 169.254.181.246
  Local 169.254.36.40 to Remote 169.254.184.165
  Local 169.254.36.40 to Remote 169.254.241.217
  Local 169.254.36.40 to Remote 169.254.93.231
Larger than PMTU communication succeeds on 12 path(s)
RPC status:
6 paths up, 0 paths down (tcp check)
6 paths up, 0 paths down (udp check)
```

Verify that the auto-revert command is enabled on all cluster LIFs	This is to make sure the interfaces failover when the switch is reloaded: <pre>network interface show -vserver Cluster -fields auto-revert</pre> If performing an RCF upgrade, please refer to those steps in this wiki article
For ONTAP 9.8 and later , enable the Ethernet switch health monitor log collection feature for collecting switch-related log files	Any issues seen at the Switch level will be logged by OnTap <pre>system switch ethernet log setup-password system switch ethernet log enable-collection</pre> <pre> CS_OTH_SY5_PRD1::*> system switch ethernet log setup-password Enter the switch name: cs_oth_sy5_prd1_sw2 (FOC24261V6C) RSA key fingerprint is cb:d4:e3:9b:68:20:b4:e3:9e:a2:80:52:f7:8b:11:be Do you want to continue? {y n}:: y Enter the password: Enter the password again: CS_OTH_SY5_PRD1::*> system switch ethernet log enable-collection Do you want to enable cluster log collection for all nodes in the cluster? {y n}: y Enabling cluster switch log collection. CS_OTH_SY5_PRD1::*> CS_OTH_SY5_PRD1::*> CS_OTH_SY5_PRD1::*> system switch ethernet log show Log Collection Enabled: true Index Switch Log Timestamp Status ----- 1 cs_oth_sy5_prd1_sw1 (FOC24261V43) - - 2 cs_oth_sy5_prd1_sw2 (FOC24261V6C) - - 2 entries were displayed. </pre>

Installing the NX-OS Software

These steps can be done on the Cluster Switch via the management network interface (there is no dependency on console access like the RCF upgrade process):

STEP	COMMAND / DETAILS
Download version of NX-OS Software	Upgrade Nexus 3000 and 3100 NX-OS Software - Cisco
The Cisco account details here can be used: Cisco Asset Recovery Process in Wiki	(or located on Citadel: \\fs-qtsennprds1\ch_netapp-managed-service\Cisco_NX-OS\)
Silence alerts and put the Cluster into maintenance mode	This needs to be done in AIQUM as well as Storage Portal: https://storage-portal.citadelgroup.com/silence Otherwise pager duty alerts will be generated and Citadel will respond.
To be performed on Switch:	
Ping the server hosting the NX-OS and the RCF to ensure connectivity	ping 10.60.127.182

Copy the NX-OS software to the Switch	copy http://10.60.127.182:8000/tmp/nx-os/nxos.9.3.8.bin bootflash: vrf management
Upload version of code from the shared location to the hosting server (specifics from the above steps, if the current HTTP server is still being used)	NGINX: copy http://10.60.127.182:82/nxos.9.3.7.bin bootflash: vrf management copy http://10.227.133.132/clusters/images/nxos.9.3.9.bin bootflash: vrf management SIMPLEHTTP: copy http://10.60.127.182:8000/tmp/nx-os/nxos.9.3.7.bin bootflash: vrf management
Show running version of the NX-OS on the switch	show version
Change directory to where the NX-OS file(s) are located	dir bootflash:
Copy running config to start	copy run start
Check for incompatibility issues	show incompatibility nxos bootflash:nxos.9.3.7.bin
Check for install impact issues	show install all impact nxos bootflash:nxos.9.3.7.bin
Install the NX-OS image	install all nxos bootflash:nxos.9.3.7.bin
To be performed on NetApp Cluster (post switch reboot):	
Confirm the status of the LIF's	net port show -ipspace Cluster
Confirm Interface Status	net int show -role Cluster -fields is-home, curr-port, home-port
To be performed on Cluster Switch (post switch reboot):	
Verify the new version of NX-OS after the switch has been rebooted	show version

IMPORTANT - You must verify node connectivity to the first switch after it is upgraded and rebooted. If the cluster interfaces to the first upgrade switch are not restored, you risk disrupting the cluster by upgrading and rebooting the second switch prematurely.

POST Check Steps to be carried out from the Cluster and the Switches

In most cases, this is the info that is needed to captured and uploaded to the IOCH as part of the PIR (post implementation review), but as previously stated, it is best practice to keep console output saved in case it is needed:

STEP	COMMAND / DETAILS																		
To be performed on Switch:																			
Confirm status of upgrade	The running version should match the version just upgraded <pre>show install all status</pre> Images will be upgraded according to following table: <table><thead><tr><th>Module</th><th>Image</th><th>Running-Version (pri:alt)</th><th>New-Version</th><th>Upg-Required</th></tr></thead><tbody><tr><td>1</td><td>nxos</td><td>7.0(3) I7 (6)</td><td>9.3 (7)</td><td>yes</td></tr><tr><td>1</td><td>bios</td><td>v08.35 (08/31/2018) : v08.34 (04/26/2018)</td><td>v08.38 (05/29/2020)</td><td>yes</td></tr></tbody></table>	Module	Image	Running-Version (pri:alt)	New-Version	Upg-Required	1	nxos	7.0(3) I7 (6)	9.3 (7)	yes	1	bios	v08.35 (08/31/2018) : v08.34 (04/26/2018)	v08.38 (05/29/2020)	yes			
Module	Image	Running-Version (pri:alt)	New-Version	Upg-Required															
1	nxos	7.0(3) I7 (6)	9.3 (7)	yes															
1	bios	v08.35 (08/31/2018) : v08.34 (04/26/2018)	v08.38 (05/29/2020)	yes															
Confirm if there have been any issues	There should be no issues reported <pre>show install all failure-reason</pre>																		
Confirm status of all interfaces	These will automatically come back online post an NX-OS or RCF upgrade <pre>show int brief grep up</pre> <pre>mgmt0 -- -- up 10.76.5.210 Eth1/7 1 eth trunk up none Eth1/8 1 eth trunk up none Eth1/9 1 eth trunk up none</pre>																		
Show current config	Similar to the pre check steps, you should see that correct NX-OS and RCF version in the banner <pre>show running config</pre> <pre>!Command: show running-config config-profile !Running configuration last done at: Mon Jan 10 16:49:33 2022 !Time: Wed Jan 12 16:57:24 2022 version 9.3(7) Bios:version 08.38</pre>																		
Confirm status of all interfaces	Confirm all interfaces from the Cluster are connected to the switch: <pre>show cdp neighbors</pre> <table><thead><tr><th>Device-ID</th><th>Local Intfrfce</th><th>Hltdtme</th><th>Capability</th><th>Platform</th><th>Port ID</th></tr></thead><tbody><tr><td>CS_OTH_SY5_PRD1-01</td><td>Eth1/7</td><td>176</td><td>H</td><td>AFF-A800</td><td>e0a</td></tr><tr><td>CS_OTH_SY5_PRD1-02</td><td>Eth1/8</td><td>137</td><td>H</td><td>AFF-A800</td><td>e0a</td></tr></tbody></table>	Device-ID	Local Intfrfce	Hltdtme	Capability	Platform	Port ID	CS_OTH_SY5_PRD1-01	Eth1/7	176	H	AFF-A800	e0a	CS_OTH_SY5_PRD1-02	Eth1/8	137	H	AFF-A800	e0a
Device-ID	Local Intfrfce	Hltdtme	Capability	Platform	Port ID														
CS_OTH_SY5_PRD1-01	Eth1/7	176	H	AFF-A800	e0a														
CS_OTH_SY5_PRD1-02	Eth1/8	137	H	AFF-A800	e0a														
To be performed on Cluster:																			

Confirm all the Cluster IC LIF's are home	These should all be set to true <pre>net int show -vserver Cluster -fields is-home</pre> <pre>(network interface show) vserver lif is-home ----- Cluster CS_OTH_SY5_PRD1-01_clus1 true Cluster CS_OTH_SY5_PRD1-01_clus2 true</pre>
Confirm all the Switch IP's respond to ping from the Cluster node(s)	Pick a different node to ping from the pre checks <pre>cluster ping-cluster -node CS_OTH_SY5_PRD1-03 -use-sitelist true</pre> <pre>CS_OTH_SY5_PRD1::*> cluster ping-cluster -node CS_OTH_SY5_PRD1-03 -use-sitelist true Host is CS_OTH_SY5_PRD1-03 Getting addresses from sitelist... Local = 169.254.93.231 169.254.157.109</pre>
Confirm Cluster can see the new NX-OS version	Make sure there are no changes post upgrade by querying the CDP output <pre>network device-discovery show -fields discovered-device -platform N3K-C3232C</pre> <pre>node protocol port discovered-device interface ----- CS_OTH_SY5_PRD1-04 cdp e0a cs_oth_sy5_prd1_sw1 (FOC24261V43) Ethernet1/10 CS_OTH_SY5_PRD1-04 cdp e1a cs_oth_sy5_prd1_sw2 (FOC24261V6C) Ethernet1/10 CS_OTH_SY5_PRD1-02 cdp e0a cs_oth_sy5_prd1_sw1 (FOC24261V43) Ethernet1/8 CS_OTH_SY5_PRD1-02 cdp e1a cs_oth_sy5_prd1_sw2 (FOC24261V6C) Ethernet1/8</pre>
Make sure LIF auto revert is re-enabled	If this was set to false as part of the NX-OS or RCF upgrade process, then it needs to be set back to true to allow the LIF's to move when there is a real issue <pre>network interface modify -vserver Cluster -lif * -auto-revert true</pre> This can be checked to make sure the LIF setting has been set <pre>network interface show -vserver Cluster -lif * -fields auto-revert</pre> <pre>CS_OTH_SY5_PRD1::*> network interface show -vserver Cluster -lif * -fields auto-revert vserver lif auto-revert ----- Cluster CS_OTH_SY5_PRD1-01_clus1 true Cluster CS_OTH_SY5_PRD1-01_clus2 true</pre>
Perform Maintenance steps	Ensure Cluster is not in maintenance mode in AIQUM and that the silence has been removed from Storage Portal: https://storage-portal.citadelgroup.com/silence

With the above commands, we have confirmed the version and connectivity to the Cluster and ISL's are back to what it was prior to the upgrade..

Applying RCF to the Cluster Switch

In most all cases, it is not necessary to upgrade a cluster RCF when performing a routine NX-OS upgrade of a Cisco CI switch - however, in certain scenarios, an RCF file upgrade will need to be completed for continued NX-OS or OnTap interoperability.

PLEASE NOTE - this process can be performed independently of an NX-OS upgrade if the requirement is only to perform an RCF upgrade, and pre an NX-OS upgrade (the recommended format unless specified by NetApp), so some of the **pre checks** performed in the NX-OS upgrade steps maybe repeated here - it is down to the individual change owner to determine if they have and whether they need to be performed or not for the RCF upgrade.

STEP	COMMAND / DETAILS
Download the RCF file to Cluster Switch or Desktop	NetApp Support Site - Cisco Ethernet Switch https://software.cisco.com/download/home/286287288/type
Perform Maintenance Steps	If we are not putting the Cluster in <i>maintenance mode</i> in AIQUM, we should send an ASUP so no cases are auto-generated <pre>system node autosupport invoke -node * -type all -message MAINT=x {where x = hours}</pre> The Cluster that the switches are connected to will need to be silenced in Storage Portal: https://storage-portal.citadelgroup.com/silence
To be performed on NetApp Cluster:	

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The N3K-C3232C reference only works for Nexus 3232c switches, if another Cluster switch is used, then that will need to be specified instead

```
system switch ethernet show
network device-discovery show -platform N3K-C3232C
```

```
CS_OTH_SY5_PRD1::> system switch ethernet show
Switch              Type              Address              Model
-----
cs_oth_sy5_prd1_sw1(FOC24261V43) cluster-network 10.76.5.210    NX3232C
  Serial Number: FOC24261V43
  Is Monitored: true
  Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(7)
  Version Source: CDP
```

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Two LIF's minimum will be dedicated to Cluster IC purposes on each node

```
network port show -role cluster
```

```
CS_OTH_SY5_PRD1::*> network port show -role cluster
Node: CS_OTH_SY5_PRD1-01
Port      IPspace      Broadcast Domain Link MTU      Speed(Mbps) Health      Ignore
-----
e0a       Cluster      Cluster          up   9000   auto/100000 healthy false
ela       Cluster      Cluster          up   9000   auto/100000 healthy false
```

3) If any are not shown as home, these may need to be checked and reverted to home

```
network interface show -role cluster -fields is-home
```

```
vserver lif is-home
-----
Cluster CS_OTH_SY5_PRD1-01_clus1 true
Cluster CS_OTH_SY5_PRD1-01_clus2 true
Cluster CS_OTH_SY5_PRD1-02_clus1 true
Cluster CS_OTH_SY5_PRD1-02_clus2 true
Cluster CS_OTH_SY5_PRD1-03_clus1 true
Cluster CS_OTH_SY5_PRD1-03_clus2 true
Cluster CS_OTH_SY5_PRD1-04_clus1 true
Cluster CS_OTH_SY5_PRD1-04_clus2 true
8 entries were displayed.
```

4) Unless manually disabled these should all be true, but **will need** to be set to **false** as part of the upgrade work

```
network interface show -vserver Cluster -fields auto-revert
```

```
vserver lif auto-revert
-----
Cluster CS_OTH_SY5_PRD1-01_clus1 true
Cluster CS_OTH_SY5_PRD1-01_clus2 true
Cluster CS_OTH_SY5_PRD1-02_clus1 true
Cluster CS_OTH_SY5_PRD1-02_clus2 true
Cluster CS_OTH_SY5_PRD1-03_clus1 true
Cluster CS_OTH_SY5_PRD1-03_clus2 true
Cluster CS_OTH_SY5_PRD1-04_clus1 true
Cluster CS_OTH_SY5_PRD1-04_clus2 true
8 entries were displayed.
```

If set to true, go ahead and disable **auto revert**

```
network interface modify -vserver Cluster -lif * -auto-revert false
```

To be performed on the Switch:

5) You will need access to gather some information before upgrade (so a ping test from the Cluster switch to the admin host is needed):

```
ping 10.60.127.182 vrf management
```

```
cs_oth_sy5_prd1_sw2# ping 10.60.127.182 vrf management
PING 10.60.127.182 (10.60.127.182): 56 data bytes
64 bytes from 10.60.127.182: icmp_seq=0 ttl=51 time=255.117 ms
64 bytes from 10.60.127.182: icmp_seq=1 ttl=51 time=255.096 ms
64 bytes from 10.60.127.182: icmp_seq=2 ttl=51 time=255.27 ms
64 bytes from 10.60.127.182: icmp_seq=3 ttl=51 time=260.913 ms
64 bytes from 10.60.127.182: icmp_seq=4 ttl=51 time=256.348 ms

--- 10.60.127.182 ping statistics ---
5 packets transmitted, 5 packets received, 0.00% packet loss
round-trip min/avg/max = 255.096/256.548/260.913 ms
```

- 6) **IMPORTANT** - please take copies of all the necessary config files before completing an RCF upgrade via console - you can save the **running config** your homedir:

```
ssh admin@[sw1_ipaddress] show running-config > {date}_{switchname1}_config.txt
ssh admin@[sw2_ipaddress] show running-config > {date}_{switchname2}_config.txt
```

```
lp-nj1-gipat001:/home/scargill $
lp-nj1-gipat001:/home/scargill $ ssh admin@10.76.5.210 show running-config > 20211230_cs_oth_sy5_prd1_sw1_config.txt
```

Example running config file: [20211230_cs_oth_sy5_prd1_sw1_config.txt](#)

You can save the **RCF file** from admin host (lp-nj1-gipat001) and save the output direct to a file on your homedir:

```
ssh admin@[sw1_ipaddress] show file {rcf.txt} > {date}_{switch}_{rcf.txt}.txt
```

```
lp-nj1-gipat001:/home/scargill $ ssh admin@10.76.5.211 show file Nexus_3232C_RCF_v1.6-Cluster-HA.txt
> 20211230_cs_oth_sy5_prd1_sw2_Nexus_3232C_RCF_v1.6-Cluster-HA.txt
```

If needed, confirm the content of the files you have saved are correct:

```
lp-nj1-gipat001:/home/scargill $ cat 20211230_cs_oth_sy5_prd1_sw1_Nexus_3232C_RCF_v1.6-Cluster-HA.txt | grep -i filename
* Filename : Nexus_3232C_RCF_v1.6-Cluster-HA.txt
lp-nj1-gipat001:/home/scargill $
lp-nj1-gipat001:/home/scargill $ cat 20211230_cs_oth_sy5_prd1_sw2_Nexus_3232C_RCF_v1.6-Cluster-HA.txt | grep -i filename
* Filename : Nexus_3232C_RCF_v1.6-Cluster-HA.txt
lp-nj1-gipat001:/home/scargill $
lp-nj1-gipat001:/home/scargill $
lp-nj1-gipat001:/home/scargill $ cat 20211230_cs_oth_sy5_prd1_sw1_config.txt | grep -i switchname
switchname cs_oth_sy5_prd1_sw1
lp-nj1-gipat001:/home/scargill $
lp-nj1-gipat001:/home/scargill $ cat 20211230_cs_oth_sy5_prd1_sw2_config.txt | grep -i switchname
switchname cs_oth_sy5_prd1_sw2
```

PLEASE NOTE - you will need to access the Cluster switches via the Raritan Console - the following is an example of how you would get access to

- Details of the Raritan Console and ports the switches are connected to (this will normally be in the IOCH):

Temporary Raritan Console access has been granted for this change as follows:

chx-dev-str-beacc01a(FOC242875H2) -> chx-dev-mgt-con03 (IP140/RU49) Port 5 - <https://chx-dev-mgt-con03>

chx-dev-str-beacc01b(FOC242875JN) -> chx-dev-mgt-con03 (IP140/RU49) Port 6 - <https://chx-dev-mgt-con03>

- Once you SSH to the IP of the Raritan Console, you will need a Zoom session with a member of the Citadel storage team who can log you into

```
Using username "admin".
admin@10.110.148.30's password:

-----
Device Type:  Dominion SX2          Model:  DSX2-48M
Device Name:  chx-qts-mgt-con01    FW Version: 2.3.0.5.3070    SN: QVY0400034
IP Address:   10.110.148.30        Idle Timeout: 0min
IP Address:   10.110.148.31
-----

Port    Port    Port Port    Port
No.     Name    Type Status Availability
-----
1 - chx-pnt-str-acc01a DCE up idle
2 - chx-pnt-str-acc01b DCE up idle
3 - chx-str-mgt-acc01a DCE up idle
4 - chx-str-mgt-acc01b DCE up idle
5 - chx-pnt-mgt-acc08a DCE up idle
6 - chx-pnt-fab-acc08a DCE up idle
7 - chx-pnt-fab-acc08b DCE up idle
8 - chx-pnt-mgt-acc08b DCE up idle
9 - chx-pnt-mgt-acc09a DCE up idle
10 - chx-pnt-fab-acc09a DCE up idle
11 - chx-pnt-fab-acc09b DCE up idle
12 - chx-pnt-mgt-acc09b DCE up idle
13 - chx-pnt-mgt-acc10a DCE up idle
14 - chx-pnt-fab-acc10a DCE up idle
15 - chx-pnt-fab-acc10b DCE up idle
16 - chx-pnt-mgt-acc10b DCE up idle
17 - cs_mkt_qts_prd1-sw1 (NETAPP) DCE up idle
18 - cs_mkt_qts_prd1-sw2 (NETAPP) DCE up idle
19 - cs_mkt_qts_prd1-sw1 (NETAPP) DCE up idle
```

- In order to access the switch console, you will need to type "port number" (and then login to the switch using the switch credentials):

```
admin > 17 _____  
You are now master for the port.  
Escape Sequence is: Control-]
```

- If you wish to come out of a console session, you need to type press " **CTRL]** " and type " quit " to get back to the main screen of console por

```
cs_mkt_qts_prd1-sw1#  
admin:cs_mkt_qts_prd1-sw1 (NETAPP):PortCommand> quit _____  
Client is disconnected.  
admin >  
admin > 18  
You are now master for the port.
```


7) On the switch being upgraded, show the ports from the command line.

PLEASE NOTE- All ports on the switch should have all the IC LIF's from each node in the Cluster connected to it, so once the ports are shutdown an example:

```
network device-discovery show -platform N3K-C3232C -port {interface connected to switch}
```

```
CS_OTH_SY5_PRD1::*> network device-discovery show -platform N3K-C3232C -port e0a
Node/      Local   Discovered
Protocol   Port    Device (LLDP: ChassisID)  Interface      Platform
-----
CS_OTH_SY5_PRD1-04/cdp
e0a        cs_oth_sy5_prd1_sw1(F0C24261V43)  Ethernet1/10   N3K-C3232C
CS_OTH_SY5_PRD1-02/cdp
e0a        cs_oth_sy5_prd1_sw1(F0C24261V43)  Ethernet1/8    N3K-C3232C
CS_OTH_SY5_PRD1-03/cdp
e0a        cs_oth_sy5_prd1_sw1(F0C24261V43)  Ethernet1/9    N3K-C3232C
CS_OTH_SY5_PRD1-01/cdp
e0a        cs_oth_sy5_prd1_sw1(F0C24261V43)  Ethernet1/7    N3K-C3232C
4 entries were displayed.
```

If you have not already taken the details of the interfaces, you should do so from the switch before shutting the interfaces down

```
show interface brief | grep up
show cdp neighbors interface Eth1/7 | grep {interface connected to switch}
```

```
cs_oth_sy5_prd1_sw1#
cs_oth_sy5_prd1_sw1# show cdp neighbors interface Eth1/7 | grep e0a
Eth1/7      131    H      AFF-A800    e0a
cs_oth_sy5_prd1_sw1# show cdp neighbors interface Eth1/8 | grep e0a
Eth1/8      146    H      AFF-A800    e0a
cs_oth_sy5_prd1_sw1# show cdp neighbors interface Eth1/9 | grep e0a
Eth1/9      161    H      AFF-A800    e0a
cs_oth_sy5_prd1_sw1# show cdp neighbors interface Eth1/10 | grep e0a
Eth1/10     170    H      AFF-A800    e0a
cs_oth_sy5_prd1_sw1#
```

Once the ports have been identified to be shutdown, go ahead and shut them down in config mode on the switch being upgraded:

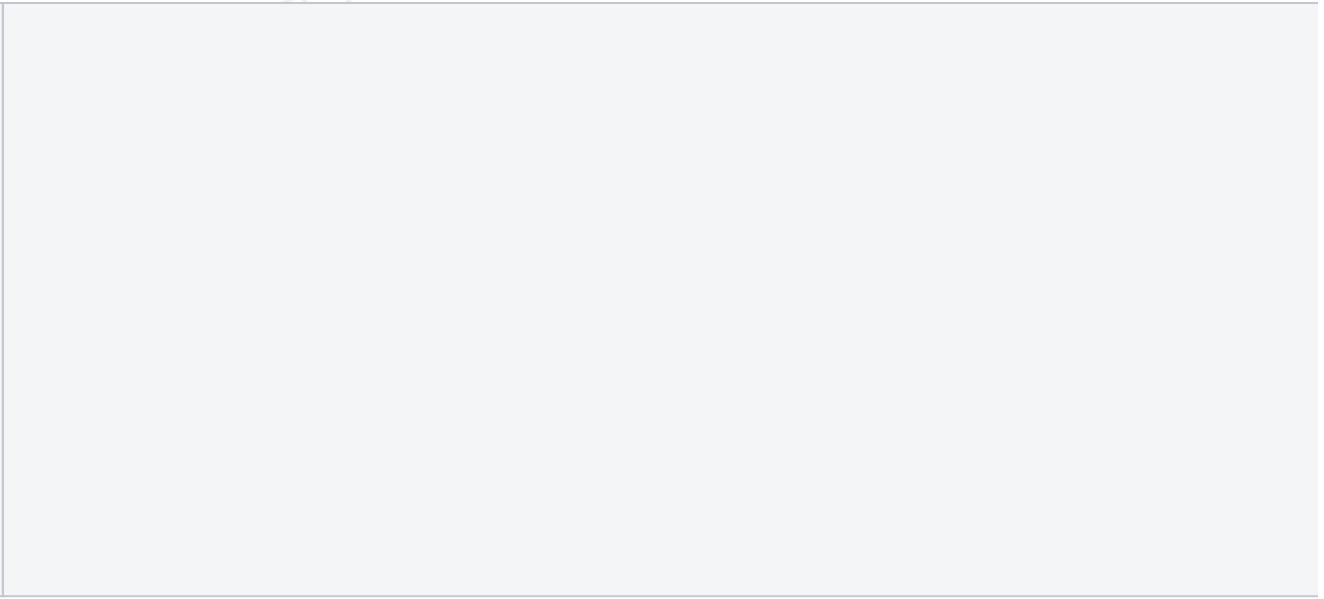
```
conf t
interface eth1/7-10
shutdown

ctrl z {to exit}
```

```
cs_oth_sy5_prd1_sw1# con t
Enter configuration commands, one per line. End with CNTL/Z.
cs_oth_sy5_prd1_sw1(config)# interface eth1/7-10
cs_oth_sy5_prd1_sw1(config-if-range)# shutdown
```

The details above are an example of the command(s) to be run and the expected output, *the actual specifics may change from Cluster to Cluster*

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8) The interfaces should be seen as **false**

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The interfaces should be seen as **false**

network interface show -role cluster

```
CS_OTH_SY5_PRD1:> network interface show -role cluster
Vserver      Logical      Status      Network      Current      Current Is
Interface    Admin/Oper   Address/Mask Node          Port         Home
-----
Cluster
CS_OTH_SY5_PRD1-01_clus1 up/up 169.254.36.40/16 CS_OTH_SY5_PRD1-01 ela false
CS_OTH_SY5_PRD1-01_clus2 up/up 169.254.25.67/16 CS_OTH_SY5_PRD1-01 ela true
CS_OTH_SY5_PRD1-02_clus1 up/up 169.254.184.165/16 CS_OTH_SY5_PRD1-02 ela false
```

9) Review any issues if the Cluster is not reporting healthy

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Review any issues if the Cluster is not reporting healthy

cluster show

```
Node          Health Eligibility
-----
CS_OTH_SY5_PRD1-01 true   true
CS_OTH_SY5_PRD1-02 true   true
CS_OTH_SY5_PRD1-03 true   true
```

To be performed on the Switch:

1) **PLEASE NOTE** - this part requires console access, so you should be logged in via the Raritan Console and NOT (for example) using SSH to the M: copied of the necessary config files.

It also good practice to have downloaded and copied the necessary RCF file to the switch prior to losing access via the management IP - for example

```
dir bootflash:
```

```
cs_oth_sy5_prd1_sw1# dir bootflash:
 4096 Oct 23 01:18:43 2021 .rpmstore/
 4096 Oct 23 01:19:04 2021 .swtam/
 1073 Aug 29 02:13:03 2020 20200829_021300_poap_31075_init.log
 1073 Aug 29 02:30:02 2020 20200829_023001_poap_31222_init.log
342639 Oct 23 00:51:14 2021 20211023_000806_poap_30885_init.log
169931 Jan 10 12:35:08 2022 20220110_122849_poap_25623_init.log
 4609 Dec 16 20:56:23 2021 Nexus-3232C-RCF-v1.8-Cluster.txt
 4289 Oct 23 01:04:23 2021 Nexus_3232C_RCF_v1.6-Cluster-HA.txt
 3188 Oct 23 01:16:59 2021 bios_daemon.dbg
```

Once you are ready, go ahead and wipe the switch:

```
write erase
```

```
reload
```

Once the switch has reloaded, perform a basic setup of the switch - the reload process may attempt to get a DHCP IP and potentially contact a TFTP that sequence by hitting **enter** and using "**skip**":

```
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_INFO: [POC428753K-BC(4A:56:DB:05:1E)] - USB disk not detected
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_INFO: [POC428753K-BC(4A:56:DB:05:1E)] - USB disk not detected (message repeated 2 times)
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_INFO: [POC428753K-BC(4A:56:DB:05:1E)] - Start DHCP v6 session
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_DHCP_DISCOVER_START: [POC428753K-BC(4A:56:DB:05:1E)] - POAP DHCP Discover phase started
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_INFO: - Abort Power On Auto Provisioning [yes - continue with normal setup, skip - bypass password and basic configuration, no - continue with Power On Auto Provisioning] (yes/skip/no): skip
[no]:
Abort Power On Auto Provisioning [yes - continue with normal setup, skip - bypass password and basic configuration, no - continue with Power On Auto Provisioning] (yes/skip/no): skip
[no]:
NOTE: You have selected skip option. POAP will be aborted and password configuration will be skipped !!!
Disabling POAP.....Disabling POAP
2022 Jan 16 00:38:48 switch 45 VDC-1 45 VPCAP-2-POAP_INFO: Rollback hard, please wait.... (This may take 3-5 minutes)
```

Once accessible you will then login to the switch using **admin** and a **blank password**, so you can start to write a minimal config - you can see from correct, so the correct IP Address and route will need to be added:

```

switch# conf t
Enter configuration commands, one per line. End with CNTL/Z.
switch(config)# do show ip int brief

IP Interface Status for VRF "default"(1)
Interface      IP Address      Interface Status
switch(config)# do show ip int brief vrf management

IP Interface Status for VRF "management"(2)
Interface      IP Address      Interface Status
mgmt0          10.76.5.127     protocol-up/link-up/admin-up
switch(config)# interface mgmt0
vrf member management
ip address 10.76.5.210/24
switch(config-if)# vrf member management
switch(config-if)# ip address 10.76.5.210/24
switch(config-if)# vrf context management
switch(config-vrf)# ip route 0.0.0.0/0 10.76.5.1
switch(config-vrf)#
switch#

```

PLEASE NOTE - we have seen in some cases, that the Cluster switch loses it's management IP on **write erase** and that's necessary to add the IP. copy of the running config prior to wiping the switch, so that we can easily write a minimal setup in terms of getting the switch visible again on the IP

```

interface mgmt0
vrf member management
ip address 10.76.5.210/24

vrf context management
ip route 0.0.0.0/0 10.76.5.1

```

These should be added in config mode:

```

conf t
vrf member management
ip address {1.2.3.4}/{gw}
vrf context management
ip route 0.0.0.0/0 {gw_ipaddress}

```

You can grep the running config to make sure this completed:

```
show running-config | grep ip
```

```

cs_oth_sy5_prdl_sw1# show running-config | grep ip
ip domain-lookup
ip route 0.0.0.0/0 10.76.5.1
ip address 10.76.5.210/24

```

However, in some other cases, the switch has maintained it management IP and **no further action** is needed in terms of adding the management IP **o determine whether action here is needed or not.**

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PLEASE NOTE - this step can be skipped if you have already copied the RCF file as part of the prep work:

copy http://10.60.127.182:82/Nexus-3232C-RCF-v1.8-Cluster.txt bootflash: vrf management

```
cs oth_sy5_prd1_sw1# copy http://10.60.127.182:82/Nexus-3232C-RCF-v1.8-Cluster.txt bootflash: vrf management
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100 4609 100 4609    0     0  9019      0 --:--:-- --:--:-- --:--:-- 9019
Copy complete, now saving to disk (please wait)...
Copy complete.
```

PLEASE NOTE - this step can be skipped if you have already copied the RCF file as part of the prep work:

dir bootflash:

```
cs_oth_sy5_prdl_sw1# dir bootflash:
 4096   Oct 23 01:18:43 2021  .rpmstore/
 4096   Oct 23 01:19:04 2021  .swtam/
 1073   Aug 29 02:13:03 2020  20200829_021300_poap_31075_init.log
 1073   Aug 29 02:30:02 2020  20200829_023001_poap_31222_init.log
342639  Oct 23 00:51:14 2021  20211023_000806_poap_30885_init.log
 4609   Dec 16 20:56:23 2021  Nexus-3232C-RCF-v1.8-Cluster.txt
 4289   Oct 23 01:04:23 2021  Nexus_3232C_RCF_v1.6-Cluster-HA.txt
 3188   Oct 23 01:16:59 2021  bios_daemon.dbg
    0    Aug 29 02:11:59 2020  bootflash_sync_list
 4096   Oct 23 01:19:06 2021  eem_snapshots/
 4096   Oct 23 01:19:02 2021  evt_log_snapshot/
  432   Dec 16 20:05:20 2021  files_list
  577   Dec 16 18:51:15 2021  id_rsa.pub
 4096   Oct 23 01:11:24 2021  lost+found/
1011037696 Aug 29 02:07:51 2020  nxos.7.0.3.I7.6.bin
1959441408 Oct 23 01:02:34 2021  nxos.9.3.7.bin
    0    Aug 29 02:13:31 2020  platform-sdk.cmd
 4096   Aug 29 02:12:35 2020  scripts/
 4096   Aug 29 02:12:47 2020  virt_strg_pool_bf_vdc_1/
 4096   Aug 29 02:12:21 2020  virtual-instance/
   59   Oct 23 02:09:25 2021  virtual-instance.conf
```

This will essentially write the config to the switch

copy Nexus-3232C-RCF-v1.8-Cluster.txt running-config echo-commands

Once it has completed, you should see the following:

```
switch(config-if)# switchport trunk allowed vlan 1
switch(config-if)# mtu 9216
switch(config-if)# no shutdown
switch(config-if)# end
switch# end
Copy complete, now saving to disk (please wait)...
Copy complete.
switch#
```

PLEASE NOTE - at this stage, the Cluster switch will have the default switch name of **switch** (as seen above) - at this time, it's good practice to rename the **running config** to the **startup config**:

```
show running-config | grep switch {equals no switch name}

conf t
switchname {switchname}

ctrl-z {to exit conf mode}

show running-config | grep switch
```

You will see the new switch name once you have set it - *in some rare cases (where a Cluster has been renamed, it may be necessary to name needed)*

- 1 You need to make sure the following information is correct (banner, node and port setting, customizations)
- 4) IMPORTANT - you need to follow the instructions of the banner **MOTD** of the RCF file being installed to ensure the RCF configuration gets applied (these steps are covered in this process - apply RCF / reload / reapply RCF):

show running-config

```
banner motd #
*****
* NetApp Reference Configuration File (RCF)
*
* Switch   : Nexus N9K-3232C
* Filename : Nexus-3232C-RCF-v1.8-Cluster.txt
* Date     : 11-01-2021
* Version  : v1.8
*
* Port Usage:
* Ports 1-30: 40/100GbE Intra-Cluster/HA Ports, int e1/1-30
* Ports 31-32: Intra-Cluster ISL Ports, int e1/31-32
* Ports 33-34: 10G Intra-Cluster/HA Ports, e1/33-34
*
* This RCF supports Clustering, HA, RDMA, and DCTCP using a single port profile.
*
* IMPORTANT NOTES
*
* - This RCF utilizes QoS and requires TCAM re-configuration, requiring RCF
  to be loaded twice with the Cluster Switch rebooted in between.
*
* - Perform the following 4 steps to ensure proper RCF installation:
*
* (1) Apply RCF first time, expect following messages:
*   - Please save config and reload the system...
*   - ERROR: Failed to write VSH commands
*   - TCAM region is not configured...
*
* (2) Save running-configuration and reboot Cluster Switch
*
* (3) After reboot, apply same RCF second time and expect following messages:
*   - % Invalid command at '^' marker
*
* (4) Save running-configuration again
*
*****
```


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This will essentially make the *rcf config* and *management ip* settings in the *running config* persistent

```
copy running-config startup-config
```

As the admin account password is "blank" post write erase, you may be unable to copy the running config file to the startup config file until you s

```
cu.  
2022 Jan 10 13:05:33 switch %SYSMGR-3-CFGWRITE_FAILED: Configuration copy failed  
(error-id 0x401E0000).  
2022 Jan 10 13:24:49 switch %SECURITYD-2-INVALID_ADMIN_PASSWD: User admin has NU  
LL password. Copy r s aborted  
2022 Jan 10 13:24:49 switch %SYSMGR-3-CFGWRITE_SRVFAILED: Service "Security Daem  
on" failed to store its configuration (error-id 0x404A0052).  
2022 Jan 10 13:24:51 switch %SYSMGR-2-CFGWRITE_ABORTED: Configuration copy abort  
ed.  
2022 Jan 10 13:24:51 switch %SYSMGR-3-CFGWRITE_FAILED: Configuration copy failed  
(error-id 0x401E0000).
```

Set the password to what it was before

```
conf t  
username admin password "password" role network-admin  
ctrl-z
```

```
switch# conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
switch(config)# username admin ?  
<CR>  
expire      Expiry date for this user account(in YYYY-MM-DD format)  
keypair     Generate SSH User Keys  
passphrase  User passphrase  
password    Password for the user  
role        Role which the user is to be assigned to  
shelltype   Choose shell type for login  
ssh-cert-dn Update cert dn  
sshkey      Update ssh key for the user for ssh authentication  
  
switch(config)# username admin password password role network-admin  
WARNING: DES is a weak encryption algorithm and currently the default. This will be deprecated and AES128 will be made default in an upcoming release.  
switch(config)#
```

PLEASE NOTE - it is important to make sure the *network-admin role* is set as part of changing the password

You can check the running config file to see that password has been added:

```
no password strength-check  
username admin password 5 $5$LAMFHB$t/ym1aNN6St2Dvy2RyjQbMLjMT8.Abk6f0BazbuOD41  
role network-admin
```

Once the password has been change, the *running config* file is now able to be copied to *startup config* file

```
switch# copy running-config startup-config  
[#####] 100%  
Copy complete, now saving to disk (please wait)...  
Copy complete.  
switch#  
switch#
```

1 6) R e b o o t t h e s w i t c h	To ensure the changes are persistent <pre>reload</pre>
1 7) A p p l y t h e R C F f i l e f o r a 2 n d t i m e	Once the Switch is back and accessible, go ahead and reapply the RCF file - this is to make sure that the details are in the TCAM database <pre>copy Nexus-3232C-RCF-v1.8-Cluster.txt running-config echo-commands</pre>
T o b e p e r f o r m e d o n N e t A p p C l u s t e r :	

The ports will be back up on the Cluster after the reload in Step 16:

```
network port show -role cluster
```

```
Node: CS_OTH_SY5_PRD1-01
Port      IPspace      Broadcast Domain Link MTU      Speed(Mbps) Health
-----
e0a      Cluster      Cluster      up  9000      auto/100000 healthy
ela      Cluster      Cluster      up  9000      auto/100000 healthy

Node: CS_OTH_SY5_PRD1-02
Port      IPspace      Broadcast Domain Link MTU      Speed(Mbps) Health
-----
e0a      Cluster      Cluster      up  9000      auto/100000 healthy
ela      Cluster      Cluster      up  9000      auto/100000 healthy
```

This may not show one switch as the LIF's are not yet homed

```
network device-discovery show -protocol cdp -platform N3K-C3232C
```

```
Node/      Local   Discovered
Protocol   Port   Device (LLDP: ChassisID) Interface Platform
-----
CS_OTH_SY5_PRD1-04/cdp
e0a        cs_oth_sy5_prd1_sw1 (FOC24261V43) Ethernet1/10 N3K-C3232C
ela        cs_oth_sy5_prd1_sw2 (FOC24261V6C) Ethernet1/10 N3K-C3232C
CS_OTH_SY5_PRD1-02/cdp
e0a        cs_oth_sy5_prd1_sw1 (FOC24261V43) Ethernet1/8  N3K-C3232C
ela        cs_oth_sy5_prd1_sw2 (FOC24261V6C) Ethernet1/8  N3K-C3232C
```

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Make sure there are no issues

cluster show

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If this is the completion of the first switch upgrade - please repeat Steps 7 to 20 on the second switch (*making sure you have a copy of the necessary files*)

If this is the completion of the second switch upgrade - please go to Step 21

To be performed on NetApp Cluster:

2.1) Enable auto-revert on the Cluster LIF's

The current status is that some of the LIF's are not home

```
net int show -vserver Cluster -lif * -fields is-home
```

```
(network interface show)
vserver lif                is-home
-----
Cluster AM_GQS_QTS_PRI-01_clus1 true
Cluster AM_GQS_QTS_PRI-01_clus2 false
Cluster AM_GQS_QTS_PRI-02_clus1 true
Cluster AM_GQS_QTS_PRI-02_clus2 false
Cluster AM_GQS_QTS_PRI-03_clus1 true
Cluster AM_GQS_QTS_PRI-03_clus2 false
Cluster AM_GQS_QTS_PRI-04_clus1 true
Cluster AM_GQS_QTS_PRI-04_clus2 false
8 entries were displayed.
```

Go ahead and make the change for the cluster LIF's to auto revert

```
network interface modify -vserver Cluster -lif * -auto-revert true
```

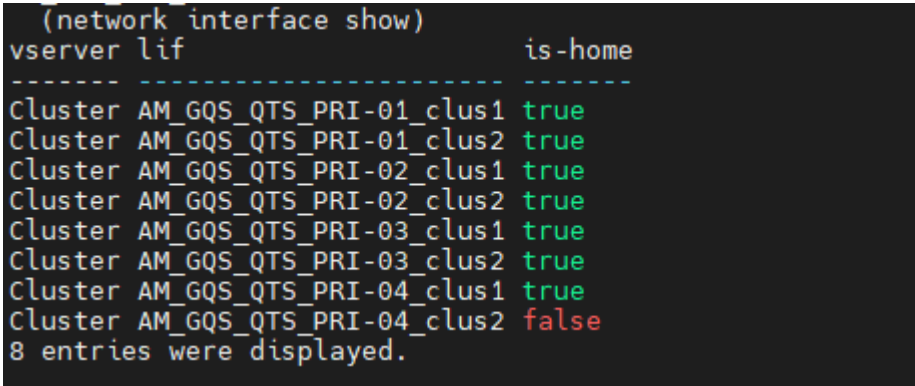
To be performed on the Switch:

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Although this can be done at the Cluster level, as we have completed an upgrade on the Cluster switch, it is good practice to make sure there are no the interfaces revert back to their home ports:

```
reload
```

Check again on the Cluster to make sure all the LIF's are home:



If they are not like the above example, you may need to force them home:

```
network interface modify -vserver Cluster -lif * -auto-revert true
```

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All the interfaces on the switches upgraded should already be online post the reload(s)

show interface brief | grep up

```
cs_oth_sy5_prd1_sw1# show interface brief | grep up
mgmt0 -- up 10.76.5.210 1000 1500
Eth1/7 1 eth trunk up none 100G(D) --
Eth1/8 1 eth trunk up none 100G(D) --
Eth1/9 1 eth trunk up none 100G(D) --
Eth1/10 1 eth trunk up none 100G(D) --
Eth1/31 1 eth trunk up none 100G(D) 1
Eth1/32 1 eth trunk up none 100G(D) 1
Po1 1 eth trunk up none a-100G(D) lacp
```

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show port-channel summary

```
cs_oth_sy5_prdl_sw1# show port-channel summary
Flags:  D - Down          P - Up in port-channel (members)
        I - Individual    H - Hot-standby (LACP only)
        s - Suspended     r - Module-removed
        b - BFD Session Wait
        S - Switched      R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met

-----
Group Port-      Type      Protocol  Member Ports
Channel
-----
1      Po1 (SU)   Eth       LACP      Eth1/31 (P)  Eth1/32 (P)
cs_oth_sy5_prdl_sw1#
cs_oth_sy5_prdl_sw1#
cs_oth_sy5_prdl_sw1#
cs_oth_sy5_prdl_sw1# show port-channel summary
Flags:  D - Down          P - Up in port-channel (members)
        I - Individual    H - Hot-standby (LACP only)
        s - Suspended     r - Module-removed
        b - BFD Session Wait
        S - Switched      R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met

-----
Group Port-      Type      Protocol  Member Ports
Channel
-----
1      Po1 (SU)   Eth       LACP      Eth1/31 (P)  Eth1/32 (P)
```

If the port channel is showing as down, check the port channel itself

show int Po1

```
port-channel1 is down (No operational members)
admin state is up,
Hardware: Port-Channel, address: bc4a.56db.0662 (bia bc4a.56db.0662)
MTU 9216 bytes, BW 100000 Kbit , DLY 10 usec
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, medium is broadcast
Port mode is trunk
auto-duplex, auto-speed
```

And the interfaces in that port channel:

show int Eth1/31

```
am_gqs_qts_pri-sw2# show int Eth1/31
Ethernet1/31 is down (suspended(no LACP PDUs))
admin state is up, Dedicated Interface
  Belongs to Po1
  Hardware: 100/1000/10000/25000/40000/100000 Ethernet, address: bc4a.56db.0662
  (bia bc4a.56db.0662)
  Description: Intra-Cluster Switch ISL Port 1/31 (port channel)
  MTU 9216 bytes, BW 100000000 Kbit , DLY 10 usec
```

You may need to wait a while for this to come back and the port channel and the interfaces within are all showing as up

PLEASE NOTE - we have seen in some cases that the interfaces in the port channel and the port channel itself remain down post an RCF upgrade the issue - this has been deemed a **transient issue** and as long as it doesn't happen again, can be safely ignored - the following is an example of th

```
Kernel uptime is 0 day(s), 0 hour(s), 46 minute(s), 23 second(s)

Last reset at 207919 usecs after Sun Jan 16 01:23:50 2022
Reason: Kernel Panic
System version: 9.3(7)
Service:
```

IMPORTANT - If we do see an issue with the ISL Port Channel, please raise a case with Cisco and get them to troubleshoot the issue - they will need

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2 All LIF's should be as before the upgrade:

network interface show -role cluster

```
CS_OTH_SY5_PRD1::*> network interface show -role cluster
Vserver      Logical   Status   Network      Current      Current Is
Interface    Admin/Oper Address/Mask Node          Port      Home
-----
Cluster
CS_OTH_SY5_PRD1-01_clus1 up/up 169.254.36.40/16 CS_OTH_SY5_PRD1-01 e0a true
CS_OTH_SY5_PRD1-01_clus2 up/up 169.254.25.67/16 CS_OTH_SY5_PRD1-01 e1a true
CS_OTH_SY5_PRD1-02_clus1 up/up 169.254.184.165/16 CS_OTH_SY5_PRD1-02 e0a true
CS_OTH_SY5_PRD1-02_clus2 up/up 169.254.155.235/16 CS_OTH_SY5_PRD1-02 e1a true
CS_OTH_SY5_PRD1-03_clus1 up/up 169.254.93.231/16 CS_OTH_SY5_PRD1-03 e0a true
CS_OTH_SY5_PRD1-03_clus2 up/up 169.254.157.109/16 CS_OTH_SY5_PRD1-03 e1a true
CS_OTH_SY5_PRD1-04_clus1 up/up 169.254.241.217/16 CS_OTH_SY5_PRD1-04 e0a true
CS_OTH_SY5_PRD1-04_clus2 up/up 169.254.181.246/16 CS_OTH_SY5_PRD1-04 e1a true
8 entries were displayed.
```

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Final check on the Cluster health in terms of connectivity

```
cluster show
```

```
CS_OTH_SY5_PRD1::*> cluster show
Node                Health Eligibility Epsilon
-----
CS_OTH_SY5_PRD1-01  true   true      true
CS_OTH_SY5_PRD1-02  true   true      false
CS_OTH_SY5_PRD1-03  true   true      false
CS_OTH_SY5_PRD1-04  true   true      false
4 entries were displayed.
```

Pick 1 or 2 random nodes to check for Cluster ping to ensure the traffic is going the via the Cluster switch to the remote node(s)

```
cluster ping-cluster -node <node name> -use-sitelist true
```

```
CS_OTH_SY5_PRD1::*> cluster ping-cluster -node CS_OTH_SY5_PRD1-01 -use-sitelist true
Host is CS_OTH_SY5_PRD1-01
Getting addresses from sitelist...
Local = 169.254.36.40 169.254.25.67
Remote = 169.254.184.165 169.254.155.235 169.254.93.231 169.254.157.109 169.254.241.217 169.254.181.246
Cluster Vserver Id = 4294967293
Ping status:
.....
Basic connectivity succeeds on 12 path(s)
Basic connectivity fails on 0 path(s)
.....
```

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Ensure Cluster is not in ***maintenance mode*** in AIQUM and the silence has been removed from Storage Portal: <https://storage-portal.citadelgroup.ci>



See Also

- [Upgrade NX3232c Cluster Switch for MetroCluster](#)
- [NetApp Certified Firmware Versions](#)
- <https://mysupport.netapp.com/site/info/cisco-ethernet-switch>